

NORMAN NWS, OK, USA

WMO: 723570

Lat: 35.1808N

Lon: 97.4378W

Elev: 360

StdP: 97.07

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 03948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-7.1	-17.9	0.8	-6.8	-13.8	1.2	-4.6	12.7	7.0	11.6	7.4	4.3	10	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	38.6	23.8	37.3	23.9	35.2	24.2	26.1	33.8	25.6	33.0	25.1	32.2	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.9	28.5	23.7	19.4	28.3	22.9	18.4	27.9	83.1	33.6	80.7	33.1	78.4	31.9	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.3	9.1	-12.9	39.8	2.5	2.6	-14.7	41.7	-16.1	43.2	-17.5	44.6	-19.3	46.5		
			-13.6	27.5	2.4	1.3	-15.3	28.4	-16.7	29.2	-18.1	29.9	-19.8	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	4.7	6.4	11.7	16.5	20.9	26.0	28.6	27.9	23.6	17.4	11.2	5.1
	DBStd	9.68	5.69	6.06	5.75	4.71	4.30	2.93	3.14	3.42	3.60	4.72	5.16	5.52
	HDD10.0	583	179	129	48	7	1	0	0	0	0	4	46	169
	HDD18.3	1794	422	335	216	87	24	0	0	0	4	77	217	412
	CDD10.0	3037	16	29	100	202	339	481	576	556	407	232	83	16
	CDD18.3	1206	0	1	9	33	105	231	317	297	161	46	5	1
	CDH23.3	13813	3	13	84	272	964	2593	4106	3685	1623	425	44	2
	CDH26.7	6758	0	2	14	59	331	1226	2299	2019	685	119	4	0
(14) Wind	WSAvg	4.3	4.3	4.5	5.0	5.2	4.7	4.4	3.8	3.5	3.5	4.0	4.4	4.3
(15) Precipitation	PrecAvg	984	39	49	71	88	134	130	91	84	93	85	48	56
	PrecMax	1429	66	119	205	201	280	296	184	207	253	260	168	151
	PrecMin	683	3	12	13	25	36	25	18	5	16	20	0	5
	PrecStd	198	23	30	49	43	64	74	51	55	55	61	49	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.0	28.8	31.4	34.2	37.8	41.1	41.3	37.1	32.3	27.3	22.5
		MCWB	12.9	15.0	16.9	19.8	22.0	25.1	24.2	23.7	22.7	20.5	17.8	14.4
	2%	DB	19.9	22.1	26.1	27.9	32.4	36.1	38.7	38.8	33.9	29.9	24.1	18.9
		MCWB	12.1	13.6	17.0	19.0	22.5	23.6	23.7	23.6	22.9	20.4	16.0	13.1
	5%	DB	17.1	18.8	23.0	26.3	30.2	33.9	37.4	37.3	32.4	27.6	22.3	17.0
		MCWB	10.6	12.7	16.0	18.0	22.0	23.5	24.0	23.8	22.5	19.2	15.7	11.5
	10%	DB	13.8	16.4	21.3	24.1	27.9	32.4	35.9	35.1	31.0	26.0	19.9	13.9
		MCWB	8.3	11.1	14.9	17.5	21.2	23.7	24.2	24.2	22.5	18.5	14.7	9.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	17.5	20.4	23.0	26.2	26.9	26.8	26.7	25.5	23.9	20.7	17.8
		MCDB	19.1	20.4	24.9	27.8	29.2	33.8	34.2	34.4	31.9	28.2	23.6	20.0
	2%	WB	13.5	15.7	19.1	21.5	24.5	25.9	25.8	25.9	24.6	22.6	18.7	15.1
		MCDB	17.6	19.8	23.5	25.9	28.9	33.3	33.8	33.9	30.7	26.8	22.6	17.3
	5%	WB	11.3	13.9	17.8	20.3	23.5	25.2	25.5	25.4	23.9	21.3	17.0	12.3
		MCDB	15.6	18.3	21.0	24.2	28.1	31.9	33.2	32.8	29.7	25.0	19.9	15.1
	10%	WB	9.0	11.4	16.0	19.2	22.4	24.4	25.0	24.7	23.2	19.9	15.4	9.6
		MCDB	14.4	15.9	20.0	23.1	27.3	30.5	32.5	31.9	28.6	24.1	18.7	13.6
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	11.7	12.2	11.7	10.9	10.8	11.3	11.5	11.8	12.2	11.9	10.8
		MCDBR	16.5	15.8	15.0	13.6	12.8	12.3	13.1	13.6	13.3	13.8	14.8	15.3
	5% WB	MCWBR	9.7	9.3	8.3	7.2	5.7	4.5	3.6	3.9	4.9	6.7	8.4	9.8
		MCWBR	14.3	13.9	11.3	11.0	10.8	10.7	11.1	11.3	10.9	10.5	12.2	12.4
(40) Clear-Sky Solar Irradiance	taub		0.295	0.300	0.341	0.376	0.413	0.424	0.425	0.433	0.387	0.346	0.314	0.292
		taud	2.506	2.507	2.416	2.332	2.298	2.312	2.340	2.259	2.366	2.476	2.501	2.518
	Ebn at Noon		915	950	931	908	874	861	857	844	870	884	883	897
		Edh at Noon	82	92	110	125	132	130	125	134	114	94	82	76
(44) All-Sky Solar Radiation	RadAvg		2.79	3.38	4.43	5.47	5.89	6.71	6.81	6.19	5.12	3.90	2.94	2.36
	RadStd		0.28	0.51	0.28	0.42	0.63	0.65	0.37	0.37	0.46	0.52	0.38	0.27

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page